

Intelligent Study Planner & Analytics Tool

Project Type: Python (CLI → GUI → Android APK via Kivy)

1. Project Overview

This project is a cleanly architected Python application designed to help students plan, track, and analyze their study behaviour. The system starts as a Command-Line Interface (CLI) application, is later extended to a desktop GUI, and ultimately packaged as an Android APK using Kivy. The core logic must remain UI-independent at all times.

2. Core Objectives

- 1 Design a maintainable and extensible Python system.
- 2 Apply software engineering principles (separation of concerns, modularity).
- 3 Perform meaningful study analytics and burnout detection.
- 4 Prepare the system for GUI and Android deployment without rewriting logic.

3. System Architecture

The system must be divided into four major layers: Models, Services, Storage, and UI. Only the UI layer may interact with the user. All computation and logic must reside in services.

4. Folder Structure (Mandatory)

```
study_planner/  
■■■ core/  
■ ■■■ models/  
■ ■■■ services/  
■ ■■■ storage/  
■ ■■■ utils/  
■■■ cli/  
■■■ gui/  
■■■ tests/  
■■■ data/  
■■■ main.py
```

5. Data Models

- 1 Course: name, weekly target hours, difficulty level.
- 2 StudySession: date, duration, focus level, mood, course reference.
- 3 Student: collection of courses and study sessions.

6. Services (Business Logic)

- 1 PlannerService: course management and session logging.

- 2 AnalyticsService: trends, averages, correlations.
- 3 BurnoutService: overwork and focus-decay detection.

7. Storage Rules

- 1 Use JSON or SQLite only.
- 2 Storage must be swappable via an interface.
- 3 No hardcoded file paths.

8. STRICT RULES (NON-NEGOTIABLE)

- 1 DO NOT use print() in core modules.
- 2 DO NOT use input() outside CLI.
- 3 DO NOT mix UI and logic.
- 4 DO NOT hardcode values or paths.
- 5 DO NOT write logic inside CLI or GUI.

9. Allowed Practices

- 1 Raise exceptions for error handling.
- 2 Return data structures from services.
- 3 Use unit tests for services and storage.

10. Weekly Development Plan

Week 1 – Foundation

- 1 Project setup and folder structure.
- 2 Implement data models.
- 3 Create storage interface and JSON storage.

Week 2 – Core Functionality

- 1 Course management.
- 2 Study session logging.
- 3 Weekly and per-course summaries.

Week 3 – Analytics & Intelligence

- 1 Trend analysis.
- 2 Burnout detection rules.
- 3 Text-based reports.

Week 4 – Polish & UI Readiness

- 1 CLI interface.
- 2 Refactoring and error handling.
- 3 Unit testing.
- 4 GUI-ready service API.

11. Final Outcome

By the end of this project, the student will have a production-quality Python application ready for desktop GUI and Android APK deployment using Kivy.